CONTRACT ADDENDUM
QUALITY ASSURANCE REQUIREMENTS

1) GOVERNMENT PROCUREMENT QA ACTIONS: INSPECTION

Government Procurement Quality Assurance (PQA) actions will be accomplished by the Government Quality Assurance Representative (QAR) at: **Contractor's Facility**

2) GOVERNMENT PROCUREMENT QA ACTIONS: ACCEPTANCE

Acceptance of supplies will be at: **Contractor's Plant**
The Government's acceptance of FATR (if required) will be at: **Destination**

3) REWORK AND REPAIR OF NONCONFORMING MATERIAL

a. Rework and Repair are defined as follows:

(1) Rework - The reprocessing of nonconforming material to make it conform completely to the drawings, specifications or contract requirements.

(2) Repair - The reprocessing of nonconforming material in accordance with approved written procedures and operations to reduce, but not completely eliminate, the nonconformance. The purpose of repair is to bring nonconforming material into a usable condition. Repair is distinguished from rework in that the item after repair still does not completely conform to all of the applicable drawings, specifications or contract requirements.

b. Rework procedures along with the associated inspection procedures shall be documented by the Contractor and submitted to the Government Quality Assurance Representative (QAR) for review prior to implementation. Rework procedures are subject to the QAR's disapproval.

c. Repair procedures shall be documented by the Contractor and submitted on a Request for Variance, DD Form 1694, to the Contracting Officer for review and written approval prior to implementation.

d. Whenever the Contractor submits a repair or rework procedure for Government review, the submission shall also include a description of the cause for the non-conformances and a description of the action taken or to be taken to prevent recurrence.

e. The rework or repair procedure shall also contain a provision for re-inspection which will take precedence over the Technical Data Package requirements and shall in addition, provide the Government assurance that the reworked or repaired items have met reprocessing requirements.

4) QUALITY POST AWARD

Post Award Conference (definition) - A first meeting of key Contractor/Government players. This conference is to assure a clear and mutual understanding of the contract between the Government and contractor. The Post Award conference includes discussions on contract terms, conditions and requirements, line items and sequence of events needed for successful execution of the subject contract effort.

The contractor shall host a post award conference/meeting at the contractor's facility to include contractor and government contracting, management, quality assurance and technical personnel no later than 60 calendar days after contract award. The contractor shall participate with the government to arrange a schedule and agenda for the post award conference prior to the meeting. The contractor shall provide the government with minutes of the post award meeting in accordance with CDRL A028~~1 (DI-ADMN-81505)~~.

5) MEASUREMENT SYSTEM EVALUATION (MSE)

(a) **Definitions.** This paragraph defines specific terms utilized throughout the rest of the clause and in the accompanying Contract Data Requirements List (CDRL) and Data Item Description (DID) (DI-QCIC-81960) in accordance with CDRL A016~~08~~. This aids in clarifying the MSE requirements to Government and contractor personnel.

(1) ***Acceptance Inspection Equipment (AIE).*** All equipment (includes AAIE defined below), special and standard, including dimensional gages, measuring equipment, test fixtures, electronic and physical test equipment, and other test equipment used for examination and test of a product to determine conformance to the Technical Data Package (TDP) which may include drawings and specifications (e.g., Detail, Performance, Weapon specifications, and QAPs).

(2) ***Automated Acceptance Inspection Equipment (AAIE).*** AIE in which the inspection and acceptance determination of the product is performed, in whole or in part, in an automatic manner.

(3) ***Contractor Inspection Equipment.*** Government-approved equipment utilized by the contractor to perform examination and tests to assure conformance to contract requirements.

(4) ***Commercial Inspection Equipment.*** Industry-developed inspection equipment of universal application, without limitations to a specific part or item, which is advertised or cataloged as available to the trade or to the public on an unrestricted basis at an established price. Examples follow:

> **(i)** ***Standard Test Equipment.*** Multi-usage equipment that is specific to a function rather than to an item. It includes such items as hardness testers, tensile strength testers, meters, weighing devices, standard gear testers, ohmmeters, voltmeters, and oscilloscopes.
>
> **(ii)** ***Standard Measuring Equipment (SME).*** Multipurpose equipment and standards used for performing measurements. It includes such items as micrometers, rulers, tapes, height gages, and protractors, etc. Standards include visual inspection equipment such as scratch and dig standards, surface finish comparator, color standards (FED-STD-595), etc.

(5) ***Nondestructive Testing.*** The development and application of technical methods to examine materials or components in ways that do not impair future usefulness and serviceability in order to detect, locate, measure and evaluate flaws; to assess integrity, properties and composition; and to measure geometrical characteristics. NDT includes Radiography/Radioscopic, Ultrasonic, Eddy Current, Magnetic Particle, and Liquid Penetrant.

(6) ***Measurement System Analysis (MSA).*** Per ASTM E2782 (Standard Guide for MSA), paragraph 3.1.7, MSA is any of a number of specialized methods useful for studying a measurement system and its properties.

(b) **Scope.** This addendum establishes requirements for design, supply, performance, and maintenance of AIE used for product inspection and acceptance. In addition, this addendum establishes requirements for the preparation, submission, and approval of AIE documentation.

(c) **AIE.** The contractor shall provide all AIE necessary to ensure conformance of components and end-items to contract requirements. AIE shall include inspection, measuring, and test equipment whether Government furnished or contractor furnished (including commercially acquired) along with the necessary specifications and procedures for their use (see ISO 10012, paragraph 6.2.1). The AIE shall not create or conceal defects on the product being inspected. All AIE documentation shall contain sufficient information to permit evaluation of the AIE's ability to test, verify, and/or measure the applicable characteristics or parameters (DI-QCIC-81960) in accordance with CDRL A016~~08~~.

(d) **AIE Designs & Government Furnished Gages.** AIE designs are of two types – Government designs (see d.1) and contractor designs (see d.2). When applicable, Government designs or Government furnished gages are designated in the TDP/contract; responsibility for all other AIE is assigned to the contractor. The designs, associated inspection procedures, and theory of operation shall have the level of detail to demonstrate capability of the proposed AIE to perform the required inspection.

(1) **Government AIE Designs.** Government AIE designs may consist of detailed drawings necessary for the fabrication and use of the AIE. Unless otherwise specified, the contractor may submit alternate or modified contractor designs of Government AIE designs.

(2) **Contractor AIE Designs.** Contractor AIE design drawings shall meet the requirements of ASME Y14.100, ASME Y14.5 and ASME Y14.43 and may include commercial inspection equipment. ["Commercial inspection equipment" is defined as shown in paragraph a.4 above. It shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data.] Designs shall be submitted for any special fixture(s) to be used. Unless otherwise specified, Gage Tolerancing Policy shall be in accordance with ASME Y14.43, "Absolute Tolerancing (Pessimistic Tolerancing)."

(3) **Visual Inspection.** Visual inspection standards used for the acceptance/rejection of product shall be submitted for approval.

(e) **AIE Package Submittals.** The contractor shall prepare the AIE package submittal in accordance with (DI-QCIC-81960) in the applicable CDRL, A016~~08~~. In addition, the contractor shall adhere to the following requirements:

(1) **Designs for Approval.** Contractor designs and/or the submission for the use of Government designs shall be approved by the Government. Partial submission of AIE designs is permissible in order to expedite the approval process; however, the response date for design review will be based on the date of the final complete submission of designs.

(2) **Correspondence in English.** The contractor shall ensure all AIE correspondence and documentation are submitted in English.

(3) **Units of Measurement.** The units of measurement within the AIE package submittal shall be consistent with the requirements of the Technical Data Package (TDP).

(4) **AIE Flow Down.** The contractor shall flow down AIE requirements to sub-contractors at any tier who are performing acceptance inspections.

(f) **Characteristics for Inspection.** AIE documentation for Critical, Special, and Major characteristic inspections shall be submitted to the Government for approval in accordance with (IAW) the CDRL A016~~08~~ (DI-QCIC-81960). AIE for Minor characteristic inspections shall be submitted to the Government for approval IAW CDRL A016~~08~~ (DI-QCIC-81960) and as required below:

☒ (1) Listed Minor (characteristics displayed on specifications and/or drawings

☐ (2) Government selected list (as attached or as provided herein)

☐ (3) Not submitted

(g) **Automated Acceptance Inspection Equipment.** The AAIE shall accept only conforming material. All characteristics requiring

AAIE per the TDP shall utilize inspection equipment with a minimum demonstrated reliability of 99.8% at a 90% confidence level to detect non-conforming material unless otherwise specified below.

(1) Reliability of 99.8% at a 90% Confidence Level for Critical/Special Characteristics

(2) Reliability of 99.0% at a 90 % Confidence Level for Major Characteristics

(3) For inspection of major and minor characteristics where contractor utilizes AAIE when it is not required by the TDP, the AAIE package shall be submitted to the Government for approval. If the Minor characteristic is not listed in paragraph f.2 or not required for submittal in paragraph f.3, then the AAIE requirements (e.g., verification, calibration, prove-out, etc.) of the inspection shall still be performed.

(4) All AAIE packages submitted to the Government for approval shall be in accordance with MIL-A-70625 (*Automated Acceptance Inspection Equipment Design, Testing and Approval of*). Furthermore, the contractor shall be responsible for producing the acceptance and rejection verification standards/masters representative of the characteristics the AAIE is designed to inspect. The verification standards and frequency of use require Government approval prior to use. When verification standards are used for the VL-VII "sampling plan" per MIL-STD-1916 paragraph 4.4, verification standards and frequency of use shall require Government approval prior to use.

(5) If the AAIE accepts a critical characteristic "reject" standard the contractor shall notify the Government and act in accordance with paragraph f of the Critical Characteristic Control Addendum. In addition, if the AAIE accepts a major and/or minor characteristic "reject" standard the contractor shall act in accordance with paragraph 8.3 of ISO 10012 or paragraph 6.4.9 of ISO/IEC 17025:2017.

(6) All AAIE shall be required to pass a Government-approved Acceptance (Prove-Out) Test. The contractor shall conduct this test per the approved test plan and shall submit a test analysis report for approval. See applicable

CDRL A016~~08~~ (DI-QCIC-81960). This test shall be performed at the contractor's facilities who's manufacturing system has had the AAIE fully integrated and calibrated as per paragraph (j) of this addendum. The contractor shall allow Government personnel access to this facility and unobstructed monitoring of this test.

(7) The contractor shall notify the Government prior to a modification and/or relocation of the Government-approved AAIE. The modified AAIE designs shall be submitted for approval. The modified and/or relocated AAIE shall require submission of the acceptance test plan (prove-out) and results for review and approval prior to use. The modified and/or relocated AAIE shall be in accordance with paragraphs (g)(1) – (g)(6).

(h) **Measurement System Analysis (MSA)**. The contractor is responsible to ensure all AIE is, at a minimum, stable, repeatable, and reproducible for all characteristics. Refer to ASTM E2782 and/or AIAG MSA for guidance. The contractor shall provide objective evidence, including the MSA assessment plan, associated data, and analysis, which demonstrates the AIE is, at a minimum, stable, repeatable, and reproducible for the following characteristics (A008 DI-QCIC-81960):

SPECIFICATION	**PARAGRAPH NO.**	**DRAWING**	**CHARACTERISTIC**
MIL-DTL-48636	4.4.2.1	13004328	M114
DTL12991155	4.4.2.3	12991157	M120
MIL-DTL-48400	4.4.2.1	11751151	M117
MIL-DTL-70679	4.4.2.1	9392366	M112
DTL12993716	4.4.2.1	12993715	M114
DTL12630554	4.4.2.3	12630579	M125
MIL-DTL-32612	4.3.3.4	13064714	M111
QAP 13064753	Part III 1.b.	13064753	M102

Approval of submitted MSA(s) must be granted before the corresponding AIE can be used or continue to be used for acceptance of product. If at any time following approval of the AIE and MSA the AIE is disapproved, then the MSA shall be disapproved. After the resubmitted AIE is approved, the MSA shall be conducted on the approved AIE and resubmitted for approval.

(i) **Robust AIE System.** The contractor shall ensure the AIE and its use is not negatively affected by any manufacturing/inspection environmental stimuli including, but not limited to production rate, noise, temperature, humidity, and vibration.

(j) **AIE Calibration and Verification.** The calibration system shall be in accordance with ISO 10012 or ISO/IEC 17025:2017. All AIE shall be subjected to scheduled calibration intervals to ensure that the equipment will accept only conforming product and reject all non-conforming product for the duration of the approved calibration period. AIE shall be subjected to periodic verification to ensure that the equipment will continue to accept and reject product with the same consistency as it did at the time of its previous calibration.

(k) **Nondestructive Testing (NDT).** Contractor shall submit detailed plans for qualifying and certifying NDT personnel and plans for qualification and ongoing use of NDT methods used for inspecting product. If re-qualification of NDT personnel and/or NDT methods is required, then the applicable plans shall be submitted.

(1) Personnel performing NDT examinations shall be qualified and certified in accordance with the standard practices prescribed by NAS 410 (NAS Certification & Qualification of NDT Personnel), ANSI/ASNT-CP-189 (ASNT Standard for Qualification and Certification of NDT Personnel), or SNT-TC-1A (Recommended Practice for Personnel Qualification and Certification in NDT), and additional procedures that may be identified by the Government. Acceptance of product using NDT shall be performed by personnel at a level of qualification consistent with that defined in the applicable standard.

(2) The NDT method(s) shall be applied in accordance with ASTM E 543 (Standard Specification for Agencies Performing Nondestructive Testing) and the current nationally recognized standard practices appropriate to the NDT method(s) employed, such as ASTM E-1742 (Standard Practice for Radiographic Examination) and SAE-AMS-STD-2154 (Inspection, Ultrasonic, Wrought Metals, Process For). Each application technique shall identify the standard(s)

utilized. Non-destructive testing includes, but is not limited to, the following types of testing: Radiography/Radioscopic, Ultrasonic, Eddy Current, Magnetic Particle, and Liquid Penetrant.

(l) **Contractor Alternate Inspection Method(s), Modifications and/or Relocation of AIE (Non-Automated) After Government Approval.** If the contractor proposes an alternate inspection method and/or modifies the AIE design(s) affecting hardware, software, or procedures after Government approval the intended change(s) shall be submitted to and approved by the Government prior to implementation. If an AIE is relocated and the relocation risks the integrity of the inspection system, notify the Government to determine information needed to assess impact to AIE. See CDRL A016~~08~~ (DI-QCIC-81960).

(m) **Responsibility for AIE Package Submittal.** The contractor shall submit the AIE design documentation package within contractual timeframes per CDRL A016~~08~~ (DI-QCIC-81960). The Government will provide approval or disapproval within the timeframe specified in the CDRL. Disapproval of the AIE package will require re-submittal and subsequent Government review in accordance with the CDRL requirements. The AIE package and any required prove-outs must be approved prior to First Article (FA) (if required) or production start-up if FA is not required.

(n) **Government's Right to Disapprove AIE.** The Government reserves the right to revoke approval of any AIE that is not satisfying the required acceptance criteria at any time during the performance of this contract. See CDRL A016~~08~~ (DI-QCIC-81960).

(o) **Navy Furnished Gages.** When gages are listed in paragraph o.9 below, the Navy Special Interface Gage (NSIG) Requirement paragraphs o.1 – o.8 shall be satisfied.

(1) The NSIG(s) are provided for verification of selected interface dimensions and do not constitute sole acceptance criteria of production items or relieve the contractor of meeting all drawing/specification requirements under the contract.

(2) The contractor is responsible for contacting the Naval Surface Warfare Center (NSWC), Corona Division at least 45 days prior to FAT (if required) or production, for the delivery of NSIG(s).

(3) NSIG(s) will be forwarded to the contractor for joint use by the Government and the contractor. Government furnished NSIG(s) shall not be used by the contractor(s) or subcontractor(s) as in-process or working gage(s).

(4) For production items that fail to be accepted by the applicable NSIG(s), an alternate inspection method may be submitted for approval.

(5) The contractor may substitute contractor designed and built AIE for the NSIG(s) noted in paragraph (o.9) below. However, the designs require Government (Navy) approval and the contractor AIE hardware requires Government (Navy) certification. AIE designs shall be submitted in accordance with CDRL A016~~08~~ (DI-QCIC-81960).

(6) The Government (Navy) shall not be responsible for discrepancies or delays in production items resulting through misuse, damage or excessive wear to the NSIG(s).

(7) Calibration and repair of the NSIG(s) shall only be performed as authorized by the NSWC Corona Division. Repair is at no cost to the contractor unless repair is required due to damage to the gages resulting from contractor fault or negligence. Damaged, worn, or otherwise unserviceable NSIG(s) shall be brought to the immediate attention of the CAO and NSWC Corona Division. The contractor shall not make any adjustments, alterations or add permanent markings to NSIG(s) hardware unless specified by the NSIG operating instructions or authorized by the NSWC Corona Division.

(8) Within 45 calendar days after final acceptance of all production items, the NSIG(s) shall be shipped to NSWC Corona Division, ATTN: Receiving Officer, Bldg 575, Gage Laboratory, 1999 Fourth St., Norco, CA 92860-1915. The following shipping and marking specifications are applicable:

(i) Shipping, MIL-STD-2073, "DOD Standard Practice for Military Packaging"

(ii) Marking, MIL-STD-129, "Marking for Shipment and Storage".

(9) The following NSIG(s) shall be provided and are mandatory for use except as noted by paragraph (o.5) above.

DRAWING	REV	CHARACTERISTIC	NSIG #	QTY	DIMENSIONS	WEIGHT	VALUE
N/A							

6) FIRST ARTICLE TEST (GOVERNMENT TESTING) a. The first article shall be examined and tested in accordance with contract requirements, the item specification(s), Quality Assurance Provisions (QAPs) and all drawings listed in the Technical Data Package.

b. The first article shall be delivered by the Contractor Free on Board (FOB) destination except when transportation protective service or transportation security is required by other provision of this contract. If such is the case, the first article shall be delivered FOB origin and shipped on Government Bill of Lading.

c. The first article shall be representative of items to be manufactured using the same processes and procedures as contract production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used during regular production. All components, subassemblies, and assemblies in the first article sample shall have been produced by the Contractor (including subcontractors) using the technical data package provided by the Government.

d. Prior to delivery, each of the first article assemblies, subassemblies, and components shall be inspected by the Contractor for all contract, drawing, QAP and specification requirements except for any environmental or destructive tests indicated here:

N/A		

The Contractor shall provide to the Contracting Officer at least 15 calendar days advance notice of the schedule date for final inspection of the first article. Those inspections which are of a destructive nature shall be performed upon additional sample parts selected from the same lot(s) or batch(es) from which the first article was selected. Results of contractor inspections (including supplier's and vendor's inspection records when applicable) shall be verified by the Government Quality Assurance Representative (QAR). One copy of the contractor's inspection report with evidence of the QAR's verification shall be forwarded with the first article; two copies shall be provided to the Contracting Officer. Upon delivery to the Government, the first article may be subjected to inspection for all contract, drawing, specification, and QAP requirements.

e. Notwithstanding the provisions for waiver of first article, an additional first article sample or portion thereof, may be ordered by the Contracting Officer in writing when (i) a major change is made to the technical data, (ii) whenever there is a lapse in production for a period in excess of 90 days, or (iii) whenever a change occurs in the place of performance, manufacturing process, material used, drawing, specification or source supply. When conditions (i), (ii), or (iii) above occurs, the Contractor shall notify the Contracting Officer so that a determination can be made concerning the need for an additional first article sample or portion thereof, and instructions provided concerning the submission, inspection and notification of results. Costs of the additional first article testing resulting from any of the causes listed herein that were instituted by the contractor and not due to changes directed by the Government shall be borne by the Contractor.

f. Rejected first articles or portions thereof not destroyed during inspection and testing will be held at the government first article test site for a period of 30 days following the date of notification of rejection, pending receipt of instructions from the Contractor for the disposition of the rejected material. The Contractor agrees that failure to furnish such instructions within said 30 day period shall constitute abandonment of said material by the Contractor and shall confer upon the Government the right to destroy or otherwise dispose of the rejected items at the discretion of the Government without liability to the Contractor by reason of such destruction or disposition.

7) FIRST ARTICLE TEST (CONTRACTOR TESTING)

a. The first article shall be examined and tested in accordance with contract requirements, the item specification(s), Quality Assurance Provisions (QAPs) and all drawings listed in the Technical Data Package.

b. The first article shall be representative of items to be manufactured using the same processes and procedures and at the same facility as contract production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used during regular production. All components, subassemblies, and assemblies in the first article sample shall have been produced by the Contractor (including subcontractors) using the technical data package applicable to this procurement.

c. The first article shall be inspected and tested by the contractor for all requirements of the drawing(s), the QAPs, and specification(s) referenced thereon, except for:

(1) Inspections and tests contained in material specifications provided that the required inspection and tests have been performed previously and certificates of conformance are submitted with the First Article Test Report.

(2) Inspections and tests for Military Standard (MS) components and parts provided that inspection and tests have been performed previously and certifications for the components and parts are submitted with the First Article Test Report.

(3) Corrosion resistance tests over 10 days in length provided that a test specimen or sample representing the same process has successfully passed the same test within 30 days prior to processing the first article, and results of the tests are submitted with the First Article Test Report.

(4) Life cycle tests over 10 days in length provided that the same or similar items manufactured using the same processes have successfully passed the same test within 1 year prior to processing the first article and results of the tests are submitted with the First Article Test Report.

(5) Onetime qualification tests, which are defined as a one-time on the drawing(s), provided that the same or similar item manufactured using the same processes has successfully passed the tests, and results of the test are on file at the contractor's facility and certifications are submitted with the First Article Test Report.

d. Those inspections which are of a destructive nature shall be performed upon additional sample parts selected from the same lot(s) or batch(es) from which the first article was selected.

e. A First Article Test Report shall be compiled by the contractor documenting the results of all inspections and tests (including supplier's and vendor's inspection records and certifications, when applicable). The First Article Test Report shall include actual inspection and test results to include all measurements, recorded test data, and certifications (if applicable) keyed to each drawing, specification and QAP requirement and identified by each individual QAP characteristic, drawing/specification characteristic and unlisted characteristic. Evidence of the QAR's verification will be provided. One copy of the First Article Test Report will be furnished to the appropriate distribution within CDRL A008~~6~~ and the Base Contract.

f. Notwithstanding the provisions for waiver of first article, an additional first article sample or portion thereof, may be ordered by the Contracting Officer in writing when (i) a major change is made to the technical data, (ii) whenever there is a lapse in production for a period in excess of 90 days, or (iii) whenever a change occurs in place of performance, manufacturing process, material used, drawing, specification or source of supply. When conditions (i), (ii), or (iii) above occurs, the Contractor shall notify the Contracting Officer so that a determination can be made concerning the need for the additional first article sample or portion thereof, and instructions provided concerning the submission, inspection, and notification of results. Costs of the additional first article testing resulting from any of the causes listed herein that were instituted by the contractor and not due to changes directed by the Government shall be borne by the Contractor.

8) DESTRUCTIVE TESTING

a. All costs for destructive testing by the Contractor and items destroyed by the Government are considered as being included in the contract unit price.

b. Where destructive testing of items or components thereof is required by contract or specification, the number of items or components required to be destructively tested, whether destructively tested or not, shall be in addition to the quantity to be delivered to the Government as set forth in the Contract Schedule.

c. All pieces of the complete First Article shall be considered as destructively tested items unless specifically exempted by other provisions of this contract.

d. The Contractor shall not reuse any components from items used in a destructive test during First Article, lot acceptance or in process testing, unless specifically authorized by the Contracting Officer.

e. The Government reserves the right to take title to all or any items or components described above. The Government may take title to all or any items or components upon notice to the Contractor. The items or components of items to which the Government takes title shall be shipped in accordance with the Contracting Officer's instructions. Those items and components to which the Government does not obtain title shall be rendered inoperable and disposed of as scrap by the Contractor.

9 USE OF MIL-STD 1916

a. The Government will not accept lots whose samples submitted for acceptance contain nonconformances unless appropriately documented and approved by the contracting officer. The contractor shall use MIL-STD 1916, DOD Preferred Methods of Acceptance of Product. The Verification Level (VL) shall be:

VL-IV for major characteristics and

VL-II for minor characteristics.

b. MIL-STD HDBK-1916 provides guidance on the use of MIL-STD 1916. This handbook is not contractually binding.

10) CRITICAL CHARACTERISTICS (SIX SIGMA)

a. The contractor's processes shall be designed with the objective of preventing the creation or occurrence of nonconforming critical characteristics (see paragraphs d & e). The contractor shall establish, document and maintain a product specific, critical characteristics control (CCC) plan that shall be submitted to and approved by the Procuring Contracting Officer (PCO) IAW , CDRL A018~~2~~ (DI-MGMT-81986). The CCC plan shall include or reference all procedures, work and handling instructions and process controls relating to any critical characteristics. Mistake Proofing techniques of the material handling and inspection systems shall be a part of the CCC Plan.

Guidance for developing this plan and submitting Critical Plans of Action (CPOA) (paragraph g) can be found at http://www.pica.army.mil/PicatinnyPublic/organizations/ardec/orgchart/quality.html.

b. The contractor shall assure its critical processes are robust in design, capable and under control, with the objective of not generating any critical non-conformances. The contractor shall calculate, document, clearly identify, and have a schedule that routinely assess the reliability and effectiveness of its critical processes to prevent generating critical non-conformances as identified in the CCC Plan.

c. An inspection and verification system shall be employed that will verify the robustness of all critical processes.
The contractor shall calculate, document, clearly identify, and have a schedule that routinely assess the reliability and effectiveness of its inspection and verification system to detect and prevent critical non-conformance escapes as identified in the CCC Plan. The Government expects that a contractor will allow zero critical escapes. To demonstrate its critical escape risk the contractor will utilize the non-conformance escape risk goal provided below.

(1) Unless otherwise specified immediately below, the calculated critical non-conformance escape risk is 1 in a million (.000001) items delivered. Or: Alternate calculated Critical Non-conformance Escape risk (fill-in -1-): Unless otherwise approved by the PCO, the non-conformance escape risk is the sum of the individual characteristic escape rates. The probability of escape for a single characteristic shall be calculated by multiplying the non-conformance rate(s) entering the inspection system(s) by the error rate of the inspection system(s). These escape rates are then summed and shall not exceed the tolerable critical non-conformance escape risk.

(2) Within 45 days after award, the contractor can elect to submit a phased-in approach on how the nonconformance escape risk will be achieved over a period of time not to exceed 180 days from the date of first article approval, or from initiation of production when first article is not required. Submission will require approval by the Government and is subject to a technical review and analysis. Allowance for a phased-in approach will then become a part of the contract. Disapproval of the

contractor’s submission does not relieve the contractor of its obligation to comply with the terms of this addendum.

(3) Based on the maximum error rate defined for the inspection system, the contractor shall develop a test procedure to demonstrate the error rate. As part of the test plan the contractor shall include sufficient test quantities to assure 90% statistical confidence in the resultant rates unless otherwise approved by the PCO. Once established, the contractor shall have a documented schedule to routinely monitor the non-conformance and inspection system error rates to assure they do not exceed the maximum rates allotted.

d. As a result of previous practices, the government’s technical data may refer to “Critical I”, “Critical II”, and “Special” characteristics. The use of the term "critical characteristics" within this addendum includes Critical I, Critical II and Special characteristics and the use of the term "critical nonconformances" includes those nonconformances pertaining to Critical I, Critical II and Special characteristics. Unless otherwise stated in Section C, these characteristics shall be subject to all requirements of this addendum.

e. In addition to critical characteristics defined in the government’s technical data (drawings, specifications, etc.), the contractor shall also identify and document in their contractor developed technical data all known material, component, subassembly and assembly characteristics whose non-conformances would likely result in hazardous or unsafe conditions for individuals using, maintaining or depending upon the product. All additional critical characteristics identified by the contractor shall comply with the critical characteristic requirements of the technical data package, supplemented herein. The Critical Item Characteristic List (CICL) review process shall be included in the CCC Plan. The contractor's additional critical characteristics shall be classified in accordance with guidance located at https://ac.ccdc.army.mil/organizations/QESA/_files/CCCP_&_CPOA_Review_Guide_Rev-A_Amd_1_UNCLASSIFIED.PDF and shall be submitted to and approved by the PCO prior to production (DI-MGMT-81988).

f. In the event that a critical non-conformance is found anywhere in the production process, the contractor, as part of its CCC Plan, shall have procedures in place to ensure:

(1) The non-conformance is positively identified and segregated to ensure that nonconforming product does not inadvertently remain in or reenter the production process. This control shall be accomplished without affecting or impairing subsequent non-conformance analysis. Final disposition of non-conforming product shall be documented and audited for traceability.

(2) The operation that produced the non-conforming component or assembly and any other operations incorporating suspect components or assemblies are immediately stopped. (See para h. for exceptions)

(3) The government (PCO) is immediately notified of the critical non-conformance via email (A013, DI-SAFT-81985).

(4) Any suspect material is identified, segregated and suspended from any further processing and shipment.

(5) An investigation is conducted to determine the root cause of the non-conformance and the required corrective actions. An evaluation shall also be conducted with regard to suspect material to ensure that no additional critical non-conformances are present. A report of this investigation shall be submitted to the government (A013, DI-MGMT-81989). The use of the DID report shall not delay notification to the government as required in f(3) above.

(6) A request to restart manufacturing or to use any suspect material associated with the critical nonconformance is submitted to the government (A013, DI-MGMT-81989). Restart of production shall not occur until authorized by the PCO, unless previously addressed in the approved CCC Plan. The Government will respond to a restart request within 3 working days. All objective evidence of the investigations to date shall be available for review at the time of restart. Suspect material shall not be used without PCO approval.

(7) The procuring activity reserves the right to refuse acceptance of any suspect material until the root cause or reasonably likely cause of the critical non-conformance has been identified, corrective action has been fully implemented and sufficient evidence has been provided to exclude non-conforming material from the conforming population.

g. The contractor may develop alternative plans and provisions, collectively referred to as a Critical Plan of Action (CPOA), relative to government or contractor identified critical characteristics. All CPOAs are independent and shall be evaluated by the government for this contract. The CPOA and any subsequent revisions submitted IAW CDRL A019~~2~~ (DI-MGMT-8199~~9~~6) require PCO approval prior to implementation. Unless otherwise specified at time of approval, contractor shall review and evaluate CPOAs for currency and process improvements at least on an annual basis and submit results to the PCO. Unless otherwise approved by the PCO, each critical characteristic shall require a separate CPOA. If the CPOA includes other documents by reference they shall be submitted upon request. Guidance for the development of a CPOA can be found in the referenced guidance located at paragraph a., of this addendum.

h. The contractor may continue production with an approved CPOA provided that the critical non-conformance is consistent with the failure mode(s) and rates established in the CPOA. Failure to meet all CPOA requirements will require the contractor to revert back to paragraph f requirements.

i. If a critical non-conformance is discovered beyond its designated inspection point and prior to Government acceptance the contractor shall take actions specified in paragraph f above. If a critical non-conformance is discovered after Government acceptance the Government has the right to invoke the requirements of paragraph f with respect to the contractor’s remaining production under this contract.

11) PROCESS CAPABILITY, CONTROL & IMPROVEMENT REQUIREMENTS

a. The Contractor shall establish a Process Control System that includes, but is not limited to, procedures, systems and software. This Process Control System shall complement the requirements of an ISO 9001-2015 or equivalent Quality Management System as well as all contract quality requirements. Statistical Process Control (SPC), when utilized, shall be implemented in accordance with ISO 11462-1 and ANSI/ASQC B1, B2, and B3 or equivalent. A Process Control Plan (PCP), which describes actions and methods to assure production processes will be in a state of control, shall be submitted to the Government for review and acceptance as stipulated in CDRL A0007~~14~~ (DI-MGMT-80004). Demonstration of process capability in accordance with the accepted PCP shall be accomplished prior to or at first article (if required) or prior to start of production. Acceptance of product shall be contingent on verification of acceptable process capability in accordance with the accepted PCP, provided all other contractual requirements are met. The Government reserves the right to withhold acceptance of product when there is evidence of noncompliance to the PCP. Should a finding of noncompliance to the PCP be made, a corrective action plan shall be submitted to the Government.

b. Characteristics for process control are as follows:

☒ (1) Characteristics for process control are attributes or features whose variation have a significant effect on product fit, form, function, performance, service life or producibility, that require specific actions for the purpose of controlling variation. Characteristics for process control result from an in depth Government-only review and analysis as specified in Technical Data Package (TDP) documentation as required below:

☒ (1.1) Government selected list, see paragraph g below

☐ (1.2) As listed key characteristics

☐ (2) Characteristics for process control are attributes or features whose variation have a significant effect on product fit, form, function, performance, service life or producibility, that require specific actions for the purpose of controlling variation. Characteristics for process control shall be determined using an in-depth Contractor review and analysis as specified in the PCP documentation. The Government reserves the right to identify any characteristics for process control as well as any additional characteristics identified in paragraph g.

☐ (3) Characteristics for process control are features within a product, subassembly, part and process whose variation from nominal (i.e., target value) significantly impacts safety, performance in terms of customer's requirements, or final cost of a product. Special controls should be applied where the cost of variation justifies the cost of control. These shall be developed from an in depth Government-Contractor review and analysis of design as specified in paragraph g below.

c. The Contractor's analysis shall include processes and operations under the control of the prime Contractor and those under the control of sub-Contractor including subtier suppliers.

The Contractor shall create a process flow chart for the entire process (including manufacturing, inspection and material handling) and perform Process Failure Modes and Effects Analysis (PFMEA) for all processes identified on the process flow chart [If option b(3) is selected, a PFMEA and process flow chart will not be necessary]. The Contractor shall identify, define and document specific controls applicable for each process and operation that affects all characteristics required for control by this clause. The Contractor shall: (a) conduct process capability studies on all process and operation parameters affecting characteristics for process control; (b) verify that all automated inspection equipment used to validate process capability has been properly calibrated and certified; and (c) conduct Measurement System Analysis (MSA) studies on all applicable corresponding measurement systems utilized to monitor process capability.

d. The Contractor shall prepare and implement a PCP. The PCP shall be based upon and include the process flow chart, PFMEA [If option b(3) is selected, a PFMEA and process flow chart will not be necessary], process capability studies and Measurement System Analysis (MSA)for all process and operation parameters affecting characteristics for process control. For each characteristic, the PCP shall describe the entire process (including manufacturing, inspection and material handling), control methods and action plans for all out of control conditions and process capability at the stated production rates. When utilizing statistical methods, a process capability index such as Cpk shall be calculated. A characteristic for process control shall be considered to have an acceptable (and capable) process if it has a Cpk of at least 2.00 for Critical characteristics, 1.33 for all other characteristics selected for control. The Contractor shall notify the Government when the minimum process capability values (Cpk) of 2.00 for Critical characteristics and 1.33 for all other characteristics for process control, or the alternative established minimum Cpk values, are no longer being maintained.

e. In accordance with MIL-STD-1916 the Contractor may request, in writing, that alternate methods of acceptance be evaluated once the processes and applicable operation parameters have been demonstrated to be both stable and capable. Any alternate methods may not be implemented until accepted by the Contracting Officer.

f. Corrective Action Requests (CARs) and Requests For Deviations (RFDs) generated for identification of product nonconformances shall result in an evaluation of the Process Control Plan (PCP). The evaluation will consider addition of new characteristics for process control to the contractually required process control list and require implementation of actions per paragraphs (c) and (d) above with submittal to the PCO for Government acceptance. If the CARs and RFDs are related to characteristics, processes and / or operations already identified in the PCP then those actions required by paragraphs (c) and (d) will be reassessed and submitted to the PCO for Government acceptance. The Government reserves the right to withhold acceptance of product until the revised PCP is accepted by the Government.

g. If box b(1)[1.1], b(2) or b(3) are checked above, the selected characteristics and applicable tools, techniques, control methods or method of analysis to obtain these are specified as follows:

Table g(1)

Drawing No.	Paragraph/ Class.	Characteristic or Requirement	Remarks	SPC Req.
	DTL12991155	60mm HE HF-1 Shell Body		
12991157	4.4.2.3/M104	Weight	Measure & chart	Yes;
	4.4.2.3/M105	Runout of PD of forwards thread at 2 projected lengths	the identified characteristic. PFMEA, process	Xbar&R
	4.4.2.3/M106	Runout of PD of rear thread at 2 projected lengths	capability study, & MSA not required.	
	4.4.2.3/M107	Runout of ogive with bourrelet		
	4.4.2.3/M108	Runout at rear of body with bourrelet		
	4.4.2.3/M110	Diameter of bourrelet		
	4.4.2.3/M111	Diameter of obturating band groove		
	4.4.2.3/M113	Outside diameter (OD) at middle basic length on taper of rear bourrelet		
	4.4.2.3/M115	Location of forward		

		basic diameter on ogive		
	4.4.2.3/M118	Thickness of wall at		
		bourrelet		
	4.4.2.3/M125	Minor dia. of fwd thread		
	4.4.2.3/M127	Major dia. of rear thread		
	4.4.2.3/M129	Overall length excluding		
		thread boss		
	4.4.2.3/M130	Location of obturating		
		band groove shoulder		
12991155	4.4.2.5/M101	Diameter of bourrelet		
	4.4.2.5/M109	Paint thickness		
	MIL-DTL-48400	60mm HE non HF-1		
11751151	4.4.2.1/M101	Weight		
	4.4.2.1/M102	Runout of PD of nose w/		
		bourrelet and front face		
	4.4.2.1/M103	Runout of PD of nose w/		
		bourrelet and rear		
		shoulder		
	4.4.2.1/M104	Runout of ogive with		
		bourrelet		
	4.4.2.1/M105	Runout at rear of body		
		with bourrelet		
	4.4.2.1/M107	Diameter of bourrelet		
	4.4.2.1/M108	Diameter of obturating		
		ring groove		
	4.4.2.1/M110	Diameter of middle basic		
		length on rear of		
		bourrelet		
	4.4.2.1/M116	Thickness of wall in		
		entire rear taper zone		
11751150	4.4.2.3/M101	Diameter of bourrelet		
	DTL12993716	60mm FRPC Shell Body		
12993715	4.4.2.1/M101	Length of body from fwd	Measure & chart	Yes;
		face to aft face forward	the identified	Xbar&R
		of thread	characteristic.	
	4.4.2.1/M102	Major dia. of aft thread	PFMEA, process	
	4.4.2.1/M107	Outside diameter within	capability study,	
		obturator groove	characteristic.	
	4.4.2.1/M108	Runout of obt. groove	& MSA not required.	
		w/ bourrelet surface		
	4.4.2.1/M109	Runout of aft thread w/		
		bourrelet and aft face		

	4.4.2.1/M110	Runout of fwd thread w/		
		bourrelet and fwd face		
	4.4.2.1/M111	Minor dia. of rear thread		
	4.4.2.1/M115	OD at three basic		
		lengths		
	4.4.2.1/M116	Diameter of bourrelet		
	4.4.2.1/M125	Weight		
12993716	4.4.2.2/M107	Paint thickness		
	MIL-DTL-70679	60mm Smoke Shell Body		
9392366	4.4.2.1/C2	Diameter of press fit	Measure & chart	Yes;
		surface	the identified	Xbar&R
	4.4.2.1/C4	Major dia. of rear thread	characteristic.	
	4.4.2.1/M101	Weight	PFMEA, process	
	4.4.2.1/M102	Runout of press fit	capability study,	
		surface with bourrelet	& MSA not required.	
	4.4.2.1/M104	Runout of ogive with		
		bourrelet		
	4.4.2.1/M105	Runout of rear of body		
		with bourrelet		
	4.4.2.1/M106	Runout of PD of rear		
		thread w/bourrelet		
		and rear shoulder		
	4.4.2.1/M108	Diameter of obturating		
		band groove		
	4.4.2.1/M117	Length from front face		
		to rear shoulder		
	4.4.2.1/M119	Length from front face		
		to rear of press fit		
		surface		
	4.4.2.1/M121	Wall thickness at		
		obturator groove		
		(after final machining)		
	4.4.2.1/M118	Length from obturator		
		groove shoulder to rear		
		shoulder		
9395680	4.4.2.3/M101	Diameter of bourrelet		
	4.4.2.3/M105	Paint thickness		
	DTL12630554	81mm HF-1 IM/non-IM		
12630579	4.4.2.3/C4	Major dia. of rear thread		
	4.4.2.3/M104	Weight	Measure & chart	Yes;
	4.4.2.3/M105	Runout of PD of forward	the identified	Xbar&R
		(female) thread at 2	characteristic.	

		projected lengths	PFMEA, process	
	4.4.2.3/M106	Runout of PD of rear	capability study,	
		thread at 2 projected	& MSA not required.	
		lengths		
	4.4.2.3/M107	Runout of ogive with		
		bourrelet		
	4.4.2.3/M108	Runout of rear of body		
		with bourrelet		
	4.4.2.3/M109	Runout of obturating		
		band groove w/ bourrelet		
	4.4.2.3/M110	Diameter of bourrelet		
	4.4.2.3/M111	Diameter of obturating		
		band groove		
	4.4.2.3/M120	Wall thickness of zone		
		which includes bourrelet		
	4.4.2.3/M129	Minor dia. of fwd thread		
	4.4.2.3/M147	Length of forward thread		
	4.4.2.3/M148	Rear thread effective		
		length		
12630554	4.4.2.5/M110	Paint thickness		
	MIL-DTL-48636	81mm FRPC Shell Body		
13004328	4.4.2.1/M101	Length of body from fwd	Measure & chart	Yes;
		face to aft face forward	the identified	Xbar&R
		of thread	characteristic.	
	4.4.2.1/M102	Major dia. of aft thread	PFMEA, process	
	4.4.2.1/M107	Outside diameter within	capability study,	
		obturator groove	& MSA not required.	
	4.4.2.1/M108	Runout of obturator		
		groove with bourrelet		
		surface		
	4.4.2.1/M109	Runout of aft thread		
		w/ bourrelet & aft face		
	4.4.2.1/M110	Runout of forward thread		
		w/ bourrelet & fwd face		
	4.4.2.1/M111	Minor dia. of fwd thread		
	4.4.2.1/M116	Diameter of bourrelet		
	4.4.2.1/M117	OD at five basic lengths		
	4.4.2.1/M126	Weight		
13004329	4.4.2.2/M101	Paint thickness		
	MIL-DTL-32612	81mm M821A4 Shell Body		
13064714	4.3.3.4/C2	Major dia. of rear thread	Measure & chart	
	4.3.3.4/C3	Wall thickness in the	the identified	Yes;

		area between the obt. groove and shoulder	characteristic. PFMEA, process capability study, & MSA not required.	Xbar&R
	4.3.3.4/M101	Overall length excluding thread boss		
	4.3.3.4/M104	Runout of PD of rear rear thread at 2 projected lengths		
	4.3.3.4/M105	Runout of obturating band groove with surface A		
	4.3.3.4/M106	Runout of o-ring groove with surface A		
	4.3.3.4/M110	OD of cylindrical section forward of obturator groove		
	4.3.3.4/M112	OD of o-ring groove bottom		
	4.3.3.4/M113	OD of obturator groove bottom		
	4.3.3.4/M117	Length from fwd to aft face		
	4.3.3.4/M120	Length of forward external thread		
	4.3.3.4/M122	Major dia. of rear thread		
	4.3.3.4/M123	Rear thread effective length		
	4.3.3.4/M136	Wall thickness of region in rear section of aft and shoulder		
	4.3.3.4/M137	Wall thickness of region in rear section of aft taper		
	4.3.3.4/M138	Wall thickness of region in center section of aft taper		
	4.3.3.4/M139	Wall thickness of region in fwd. section of aft taper		
	4.3.3.4/M140	Wall thickness of region in rear section of ogive		
	4.3.3.4/M141	Wall thickness of region in center section of ogive		
	4.3.3.4/M142	Wall thickness of region in fwd. section of		

		ogive			
	4.3.3.4/M157	Wall thickness of region			
		immediately rear of			
		fwd. thread			
	4.3.3.4/M154	Weight			
	4.3.3.4/M155	Runout of PD of forward			
		male thread at 2			
		projected lengths			
13064713	4.3.3.6/M111	Paint thickness			

	QAP 13064753	81mm M821A4 mass sim.			
13064752	III.1.b/M102	Dia. at basic length rear			
		of obt. groove			
	III.1.b/M104	Major dia. of rear thread			
	III.1.b/M105	Projected runout of rear			
		thread (2 places)			
	III.1.b/M107	Bourrelet diameter			
	III.1.b/M111	Diameter of obt. groove			
	III.1.b/M113	Length of front face to			
		obt. groove face			
	III.1.b/M118	Length of fwd. thread			
	III.1.b/M120	Minor dia. of fwd threads			
	III.1.b/M121	Projected runout of fwd			
		threads (2 places)			
	III.1.b/M125	Weight			
	unlisted	Length of rear thread			
	unlisted	Runout of obt. groove			
		and surface A			
	unlisted	OD of cylindrical			
		section forward of			
		obturator groove			
13064753	III.1.c/M102	Paint thickness			

	MIL-STD-171				

11751150	Note 4				
9395680	Notes 4&5	Protective finish	Provide description	Yes;	
12630554	Note 3		of process control	Xbar&R	
12991155	Note 3		procedures including		
12993716	Note 4		the monitoring and		
13004329	Note 2		charting of applic-		
13064713	Note 3		able parameters.		
13064753	Note 2		(Examples: how		
			immersion times are		
			monitored and		
			controlled for each		

			bath, how often	
			purity of chemical	
			baths are monitored,	
			etc. For additional	
			information refer	
			to TT-C-490H, and	
			especially paras	
			4.7.1, 4.7.3,	
			4.7.3.1, 4.7.4,	
			4.7.4.1 & 4.7.5.1.)	
			PFMEA, process	
			capability study,	
			& MSA not required.	

12) AMMUNITION DATA CARDS (ADC)

Detailed requirements and guidance for the preparation of Ammunition Data Cards (ADCs) and Ammunition Lot Numbers are contained in MIL-STD-1168, (DI-MISC-80043B)(CDRL A021~~15~~) and the Worldwide Ammunition-data Repository Program (WARP) online user's manual. Detailed requirements for obtaining and using a manufacturer's identification symbol, which is an integral component of the ammunition lot number, can be found in MIL-STD-1168 and the WARP user’s manual. Information provided in paragraphs 6.7 through 6.16 of MIL-STD-1168 shall be considered mandatory requirements where all instances of the term "should" are considered to be replaced with the word "shall."

(a) The contractor shall develop and submit ADCs in accordance with the requirements of this addendum, MIL-STD-1168, and the user's manual located on the WARP database. The WARP application is accessed through the Munitions History Program (MHP) website. The ADC requirement is a flow-down requirement that applies to contractors and their suppliers, vendors or subcontractors.

(b) The contractor shall prepare an ADC for each lot of item(s) being produced under this contract, regardless of whether or not those lots are accepted or rejected by the Government. The ADC shall comply with MIL-STD-1168 and WARP requirements.

(c) Unless otherwise authorized by the Procuring Contracting Officer, the contractor shall include, in the components sections on the ADC representing the deliverable item, as a minimum; all assemblies, sub-assemblies, components, explosives, and propellants listed below for the item being procured.

End Item Component Listing:

Drawing Number Nomenclature

12991155 60mm HE HF-1 Shell Body
- 12991157 60mm HE HF-1 Shell Body (unpainted)
- 12991155 Material (Shell Body)
- 12991155 Phosphate protective finish
- 12991256 Primer, Interior
- 12997579 Topcoat, Ammunition
- 12995075 Bag, VCI

9395680 60mm M722/M722A1 Smoke Shell Body
- 9392366 60mm M722/M722A1 Smoke Shell Body (unpainted)
- 9392366 Material (Shell Body)
- 9395680 Phosphate protective finish
- MIL-DTL-11195 Topcoat, Ammunition
- 12995075 Bag, VCI

12993716 60mm M769 FRPC Shell Body
- 12993715 60mm M769 FRPC Shell Body (unpainted)
- 12993715 Material (Shell Body)
- 12993716 Phosphate protective finish
- MIL-DTL-11195 Topcoat, Ammunition
- 12995075 Bag, VCI

12630554-1 81mm HE HF-1 IM Shell Body
- 12630579-1 81mm HE HF-1 IM Shell Body (unpainted)
- 12630579-1 Material (Shell Body)
- 12630554-1 Phosphate protective finish
- 12991256 Primer, Interior
- 12997579 Topcoat, Ammunition
- 12995075 Bag, VCI

12630554-2 81mm HE HF-1 non-IM Shell Body
- 12630579-2 81mm HE HF-1 non-IM Shell Body (unpainted)
- 12630579-2 Material (Shell Body)
- 12630554-2 Phosphate protective finish
- 12991256 Primer, Interior
- 12997579 Topcoat, Ammunition
- 12995075 Bag, VCI

13004329 81mm M879A1 FRPC Shell Body
- 13004328 81mm M879A1 FRPC Shell Body (unpainted)
- 13004328 Material (Shell Body)
- 13004329 Phosphate protective finish
- 12991256 Primer, Interior
- MIL-DTL-11195 Topcoat, Ammunition and Identification Band
- 12995075 Bag, VCI

13064713 81mm M821A4 Shell Body
- 13064714 81mm M821A4 Mass Simulator (unpainted)
- 13064714 Material (Shell Body)
- 13064713 Phosphate protective finish
- 12991256 Primer, Interior
- 12997579 Topcoat, Ammunition
- 12995075 Bag, VCI

13064753 81mm M821A4 Mass Simulator
- 13064752 81mm M821A4 Mass Simulator (unpainted)
- 13064752 Material (Shell Body)
- 13064753 Phosphate protective finish
- MIL-DTL-11195 Topcoat, Ammunition and Identification Band
- 12995075 Bag, VCI

(d) The component items identified below are from paragraph (c) above and will require their own component ADC in addition to being listed on the end item ADC. The component ADCs shall also comply with MIL-STD-1168 and WARP requirements.
Drawing Number from paragraph (c) above, components as follows:
N/A

(e) Lot numbers shall be in accordance with MIL-STD-1168 lot number convention and the technical data package requirements. Lot numbers shall be used for all ammunition end items and their major components, including inert, dummy, or non-energetic items and components. When not required by technical data package and not considered an end item or major component, the component lot number may be constructed through contractor lot number convention.

(f) The flowdown of the requirement for component ADCs generated via WARP is highly encouraged for other items not identified in paragraph (d) above when the prime contractor is purchasing components, assemblies, and subassemblies from subcontractors or vendors.

(g) All component RFD/ECPs shall be listed on the ADC for the deliverable item, as well as on the component ADC, when that component is identified in paragraph (d) above. The WARP user's manual provides information on the level of detail required.

(h) A sample ADC shall be developed and submitted to the WARP system 30 days prior to First Article testing or 30 days prior to production in the event a first article is not required. The WARP ADC program will not allow the submission of additional ADCs until such time as the sample ADC has been approved in the system.

END OF CONTRACT ADDENDUM, QUALITY ASSURANCE REQUIREMENTS.